Problem sheet 9 for Rigid Analytic Geometry Summer Term 2026

We assume K to be algebraically closed and complete with a non-discrete ultrametic absolute value. Recall $\mathfrak{K}_{\leq R}$ and similar notations from sheet 7, as well as $B_{<R}(x) = \left\{y \in K \;\middle|\; |y-x| < R\right\}$ and similarly for $B_{\leq R}(x)$. We are only interested in these balls when $R \in (0,1]_{\mathbb{R}}$ and $x \in K^\circ$. In this case, x defines a van der Put point of $X = K^\circ = \mathrm{Sp}\mathbb{T}_1$. On the previous sheet we introduced $B_{\leq R}(\xi)$ and $B_{<R}(\xi)$ for all van der Put point and $R \in (0,1]_{\mathbb{R}}$, the only ball of the indicated type belonging to ξ. Only $B_{\leq 1}(\xi) = K^\circ$ is guaranteed to exist for arbitrary ξ. As any open subset containing $U \in \xi$ is also $\in \xi$, when $B_{<r}(\xi)$ exists then $B_{\leq r}(\xi)$ as well all $B_{<R}(\xi)$ with $R \in [r,1]_{\mathbb{R}}$ also exist (consider the appropriate kind of ball around any element of $B_{<r}(\xi)$) . If $B_{\leq r}(\xi)$ exists and $R \in [r,1]_{\mathbb{R}}$ then $B_{\leq R}(\xi)$ also exists. The infimum of r for which $B_{\leq r}(\xi)$ exists is denoted R_ξ as on the previous sheet. By the above observations and the density of $|K^\times|$ in $(0,\infty)_{\mathbb{R}}$, the value of R_ξ would not change if only $r \in |K^\times| \cap (0,1]_{\mathbb{R}}$ are considered. Moreover, safe for the case $R_\xi = 1$ one could consider the $< r$-balls instead of the $\leq r$-balls without changing R_ξ.

On the previous sheet, the case $R_\xi = 0$ was found to give precisely the van der Put points given by classical points, i. e., by $x \in K^\circ$. The case $R_\xi \notin |K|$ is mostly easy to settle by using results from the previous sheets, provided that $\bigcap_{R \in (R_\xi, 1]} B_{\leq R}(\xi)$ is not empty.

Problem 1 (1 point)**.** *Let $\xi \in X^*$ with $R_\xi \in (0,1)_{\mathbb{R}} \setminus |K^\times|$ such that $B_{\leq r}(0) \in \xi$ for all $r \in (R_\xi, 1]_{\mathbb{R}}$. Show that ξ coincides with the van der Put point from Problem 3 on sheet 7.*

Remark 1. *The case where $R_\xi \in (0,1)_{\mathbb{R}} \setminus |K^\times|$ and the intersection $\bigcap_{R \in (R_\xi, 1]} B_{\leq R}(\xi)$ is not empty is related to this case by translation by any x belonging to that intersection. However, the intersection can easily be empty for certain K, even for fairly natural ones like $K = \mathbb{C}_p$.*

For the purposes of the following definition and the problem following it, it is best to briefly drop the assumption that K be algebraically closed.

Definition 1. *Let K be any field equipped with a non-discrete absolute value. We say that K is* spherically complete *if any set $\mathfrak{L}$ of balls in K which is linearly ordered by $\subseteq$ has non-empty intersection in K.*

Problem 2 (2 points)**.** *If K is spherically complete then K is complete.*

Problem 3 (2 points)**.** *If $|K^\times|$ is a discrete subset of $(0,\infty)_{\mathbb{R}}$, then K is spherically complete if and only if it is complete.*

Problem 4 (4 points)**.** *If $|K^\times|$ is dense in $(0,\infty)_\mathbb{R}$ and K has a countable dense subset, then K cannot be spherically complete.*

For this reason, $\mathbb{C}_p$ is spherically incomplete.

Definition 2. *Assume that $|K^\times|$ is dense in $(0,\infty)_\mathbb{R}$. A* mock circle[1] *of radius r is a $\subseteq$-linearly ordered set $\mathfrak{M}$ of $\leq$-balls in K such that $\mathfrak{M} \cap \mathfrak{K}_{\leq R} \neq \emptyset$ for all $R \in (r,\infty)_\mathbb{R}$ and such that the intersection of the elements of $\mathfrak{M}$ is empty.*

We now restore our assumption that K be algebraically closed. In the lecture it will be shown that every non-zero element f of $\mathbb{T}_1$ is multiplicatively equivalent to some polynomial. Thus, f has only finitely many zeroes in K°. Using this and Problem 3 from sheet 4, one gets

Problem 5 (2 points)**.** *If $f \in \mathbb{T}_1$ and $\mathfrak{M}$ is a mock circle in K such that $K^\circ \in \mathfrak{M}$, then there is $B \in \mathfrak{M}$ such that $x \to |f(x)|$ is constant on B.*

Let $\|f\|_\mathfrak{M}$ be $|f(x)|$ for any $x \in B$, with B as above. Using the fact the $\mathfrak{M}$ is $\subseteq$-linearly ordered, one easily sees that this does not depend on the choice of B.

Problem 6 (5 points)**.** *Let $\mathfrak{M}$ be as in the previous problem, let $R \in (0,1)_\mathbb{R}$ be the radius of the mock circle $\mathfrak{M}$, let $(f_i)_{i=0}^m \in \mathbb{T}_1^{m+1}$ such that $\mathbb{T}_1 = \langle f_0, \dots, f_m \rangle_{\mathbb{T}_1}$, and let $\Omega = \mathcal{R}_X(f_0|f_1, \dots, f_m) \subseteq K^\circ$.*

- *If $\|f_i\|_\mathfrak{M} \leq \|f_0\|_\mathfrak{M}$ for all $i \in [1,m]_\mathbb{N}$ then there is $B \in \mathfrak{M}$ such that $B \subseteq \Omega$.*
- *If there is $i \in [1,m]_\mathbb{N}$ such that $\|f_i\|_\mathfrak{M} > \|f_0\|_\mathfrak{M}$ then there is $B \in \mathfrak{M}$ such that $B \cap \Omega = \emptyset$.*

Moreover, there is a unique van der Put point $\xi_\mathfrak{M} \in X^*$ such that $\xi_\mathfrak{M} \cap \mathfrak{Rat}_X$ is precisely the set of rational open subsets Ω for which the first point in the above alternative occurs.

Problem 7 (2 points)**.** *Let $\xi \in X^*$ be such that $R_\xi \in (0,1)_\mathbb{R}$ and $\bigcap_{r \in (R_\xi, 1]_\mathbb{R}} B_{\leq R}(\xi) = \emptyset$. Let $\mathfrak{M}$ be the mock circle of radius R_ξ containing all $B_{\leq R}(\xi)$ with $R \in (R_\xi, 1]_\mathbb{R}$ and all $B_{\leq R}(0)$ with $R \in (0,\infty)_\mathbb{R}$. Show that $\xi = \xi_\mathfrak{M}$.*

It remains to study the van der Put points with $R_\xi \in |K^\times|$ and such that $B = \bigcap_{r \in (R_\xi, 1]_\mathbb{R}} B_{\leq r}(\xi) \neq \emptyset$. This splits into the two cases where

[1] I am not aware of a generally used notion for this phenomenon. One could also think of $\mathfrak{M}$ as a "black hole" of radius r. In the case where $K^\circ \in \mathfrak{M}$, the valuations of $\mathbb{T}_1$ corresponding to such $\mathfrak{M}$ are the ones called "dead ends" in P. Scholze, Perfectoid Spaces, Example 2.20.

$B \notin \xi$ and $B \in \xi$, and shall see on the next sheet that the second of these cases splits into two subcases. The case $B \notin \xi$ is linked to the valuation N_r with $r \in (0,1)_{\mathbb{R}} \cap |K^\times|$ studied in subsection 1.1 in the very first lecture of this course.

Let $r \in (0,1)_{\mathbb{R}} \cap K^\times$. Recall

$$N_r(f) = \begin{cases} 0 & f = 0 \\ (n,j) \in (0,\infty)_{\mathbb{R}} \times \mathbb{Z} & f \neq 0. \end{cases} \tag{1}$$

where in the second case

$$n = \max_{0 \leq j \leq \infty} r^j \, |f_j| = \max\big\{ |f(x)| \;\big|\; x \in B_{\leq r}(0) \big\} \tag{2}$$

and j in (1) is largest with the property that $r^j \, |f_j| = n$. In (1), $(0,\infty)_{\mathbb{R}} \times \mathbb{Z}$ is ordered lexicographically with the first factor being more significant:

$$(n,j) \preceq (\nu,\iota) \text{ if and only if } n < \nu \text{ or } n = \nu \text{ and } j \leq \iota.$$

Problem 8 (3 points)**.** *Prove (1.1.2) from the first lecture in this term: For $f \in \mathbb{T}_1 \setminus \{0\}$ and (n,j) as in* (1)*, there is $R \in (r,1]_{\mathbb{R}}$ such that*

$$|f(x)| = n\Big(\frac{|x|}{r}\Big)^j$$

when $r < |x| < R$.

Remark 2. *Note that* (1) *can be applied for arbitrary $r \in (0,1]_{\mathbb{R}}$, safe for the fact that in the case $r \notin |K^\times|$ the second* max *in* (2) *must be replaced by a* sup*. The subgroup $\Gamma_{N_r} \subseteq (0,\infty)_{\mathbb{R}} \times \mathbb{Z}$ generated by the values of N_r is $\Gamma_{N_r} = |K^\times| \times \mathbb{Z}$ when $r \in |K^\times|$ while in the case $r \notin |K^\times|$, the projection of $(0,\infty) \times \mathbb{Z}$ to the first factor is an order-preserving isomorphism of Γ_{N_r} onto the subgroup $r^{\mathbb{Z}} \cdot |K^\times| \subseteq (0,\infty)_{\mathbb{R}}$. In this case N_r is equivalent to the valuation from Problems 5 and 6 from sheet 6. Also, the equivalence class of N_1 is the only element of* $\mathrm{Cont}\mathbb{T}_1 \setminus \mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$*. It is not related to any element of X^*.*

Problem 9 (2 points)**.** *Let A be a nat-ring, $X \subseteq A$ a topologically nilpotent subset and $X' \subseteq A$ the subgroup of $(A,+)$ generated by X. Show that X' is topologically nilpotent!*

Of the 23 points from this sheet, three are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday June 29. Good luck!